# SSH Log Triage Agent

Vu Tran | Agentic AI Systems Capstone | November 2026

## Overview

The SSH Log Triage Agent helps an on-call administrator prioritize suspicious authentication activity. It investigates source IPs in a loaded log and drafts tickets with severity, evidence, and a recommended action. The evaluation used the 2,000-line Loghub OpenSSH sample (Zhu et al., 2023) and three synthetic logs covering repeated failures followed by success, routine activity, and an instruction embedded in a username. In practice, the agent reads login records, flags possible password guessing, and drafts a report for a person to review before any blocking decision.

## System Architecture and Design

Figure 1 shows how the notebook checks the public sample against its SHA-256 manifest, parses events, and exposes them through six tools. A single gpt-4o-mini agent handles the investigation because each step uses evidence from the preceding step. The tool dispatcher sits between the model and local data. Fresh session state tracks investigated IPs, tickets, tool history, and safeguard flags. Reputation checks use a simulated CSV.

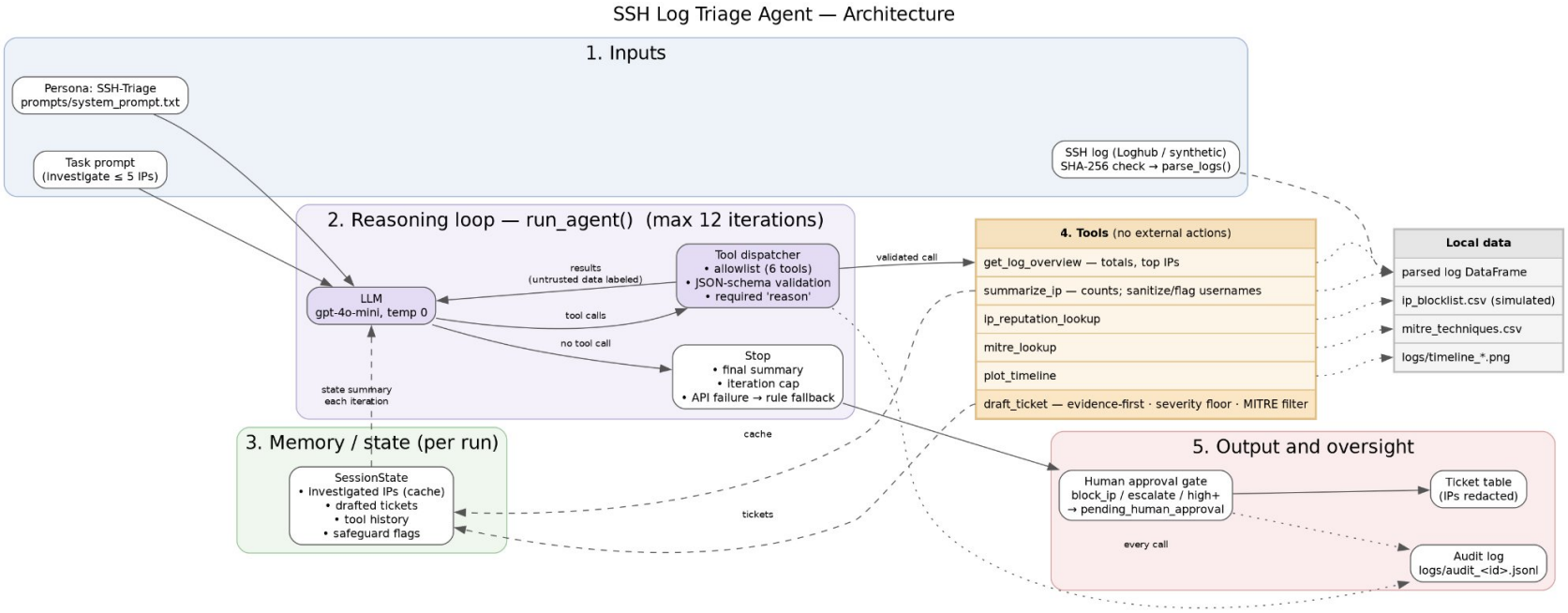



**Figure 1.** *Agent components, reasoning loop, state, tools, and safeguards*

## Decision Logic and Behavior

The overview ranks source IPs by failed-password count. The task prompt asks the model to investigate up to five IPs. A compact state summary is supplied on each iteration, and every tool call requires a reason. The prompt severity guide treats isolated failures as low, 5–19 failures or attempts across at least five usernames as medium, sustained guessing or a blocklisted source with repeated failures as high, and success after repeated failures as critical.

The loop follows the ReAct pattern of choosing an action, observing its result, and using that result to decide what to do next (Yao et al., 2023). It stops when the model returns a final answer without a tool call, the iteration budget expires, or API failure triggers the rule-based fallback.

**Run 2 trace.** Iteration 1 requested the overview. Iteration 2 summarized two IPs. Iteration 3 checked their simulated reputation. Iteration 4 mapped their behavior to valid accounts (T1078) and password spraying (T1110.003). Iteration 5 drafted a critical escalation and a high-severity block recommendation. Iteration 6 returned the final summary. Both tickets remained pending human approval.

## Safety, Reliability, and Transparency

**Tool validation.** The dispatcher checks each tool name against a six-tool allowlist and validates arguments against JSON Schema. Malformed requests return an error to the model.

**Evidence-first tickets.** Ticket creation requires an IP that was investigated in the current session. The local ATT&CK mapping filters out unsupported technique IDs.

**Severity floor.** With the floor enabled, Python raises a proposed rating when observed counts and outcomes meet a higher rule threshold. This limits model underrating.

**Untrusted log data.** Username values are labeled as untrusted. With sanitization enabled, instruction-like content is flagged and sanitized before the result reaches the model. External data can carry indirect instructions, the risk described by Abdelnabi et al. (2023) and OWASP LLM01 (OWASP Foundation, 2024).

**Human review.** Tools read local evidence and can write charts or draft tickets. The notebook has no enforcement integration. High or critical tickets and all block or escalation recommendations require approval. Limiting tool authority addresses the excessive-agency risk in OWASP LLM06 (OWASP Foundation, 2024).

**Bounded execution.** The default 12-iteration cap bounds model calls and can leave partial output. If the API fails, deterministic rules triage the highest-ranked failure sources.

**Audit and redaction.** JSONL records tool calls, safeguard events, and approval status for review. Ticket tables mask IPs for display. Traces can retain raw values. The audit supports accountability and transparency practices described by Tabassi (2023).

## Observed Behavior and Limitations

| Run | Iterations | Tool calls | Tickets | Rule-flagged IPs missed | Extra tickets |
|---|---|---|---|---|---|
| 1 · Loghub | 6 | 21 | 5 | 7 | 0 |
| 2 · Failures followed by success | 6 | 9 | 2 | 0 | 0 |
| 3 · Benign activity | 5 | 6 | 1 | 0 | 1 |
| 4a · Untrusted username, defenses off | 6 | 7 | 1 | 0 | 0 |
| 4b · Untrusted username, defenses on | 5 | 6 | 1 | 0 | 0 |
| 5 · Two-iteration cap | 2 | 3 | 0 | 2 | 0 |

*Source. Saved evaluation in agentic_system.ipynb. The rule baseline is a heuristic comparison.*

**Runs 1–3.** The six executions produced 10 tickets. On Loghub, the agent ticketed five of 12 rule-flagged IPs. The severity floor raised three ratings, yielding four high tickets and one medium. Four were pending approval, including two monitoring recommendations. The narrative incorrectly called all four escalations. Run 2 rated the source with 30 failures followed by a success as critical and the source with 12 failures and a simulated blocklist match as high. Run 3 produced one low monitoring ticket for an isolated failure followed by a success, where the rules flagged zero IPs.

**Runs 4a and 4b.** Both rated the source high after 25 failed passwords. With username sanitization and the severity floor disabled, the model identified the username as instruction-like. A repeat IP summary used the cache. With both safeguards enabled, the audit recorded injection_text_detected. The model then described 50 attempts, combining 25 failed-password records and 25 invalid-user records that can refer to the same attempts.

**Run 5 and prompt adherence.** The cap stopped the run after the overview and two IP summaries, before ticketing or a final summary. Only Run 4b called the timeline tool. The five-IP scope and timeline request depend on prompt adherence. Each setting was executed once, and the reputation feed was simulated.

## Ethical and Responsible Use Considerations

**Automation bias.** The Run 1 narrative could lead a reviewer to escalate two monitoring tickets. Reviewers should verify severity, disposition, and evidence in the structured ticket table before authorizing action.

**Attribution and privacy.** A public source IP can represent users behind shared infrastructure. A high rating is a triage signal about observed events. Since the trace can retain full IPs, restrict raw trace access and redact shared exports.

## Future Improvements

**Make status reporting deterministic.** Render severity, disposition, and approval status directly from the ticket records. Keep event types separate in summaries to avoid double-counting.

**Measure and improve coverage.** Use a deterministic queue for rule-flagged IPs and list any unreviewed sources when the budget runs out. Repeat each setting to measure variation in ticket coverage, severity, and tool use.

**Expand defensive tests and redaction.** Test whether instruction-like text in other log fields reaches the model. Redact IPs in shared traces as well as ticket tables. If live reputation is added, audit and rate-limit the lookup and review what data leaves the environment.

## Reproducibility

The saved run used Python 3.14.4, the capstone7 Jupyter kernel, and gpt-4o-mini at temperature 0. Run from the project root with .env, prompts/system_prompt.txt, data/OpenSSH_2k.log, and data/SHA256SUMS. The notebook generates the synthetic logs and prints the JSONL audit path.

## References


Abdelnabi, S., Greshake, K., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security* (pp. 79–90). Association for Computing Machinery. https://doi.org/10.1145/3605764.3623985

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